Retail Sales Analysis & Data Storytelling

Data Analytics Internship Project

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Project Overview

Objective

Analyse retail sales data to uncover actionable business insights, identify performance trends, and support strategic decision-making across categories, regions, and time periods.

Explore

Identify patterns in sales, region, and category data

Visualise

Build an interactive Power BI dashboard

Recommend

Deliver data-driven business recommendations

Dataset Overview

9,994

Total Rows

Retail transaction records

16

Columns

Features per record

Key Columns in the Dataset

Column	Description
Order Date	Date of each transaction
Category	Product category (e.g., Technology)
Region	Geographic sales region
Quantity	Units sold per order
Sales	Revenue generated per order

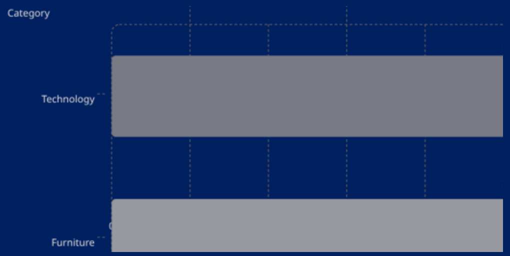
Data Cleaning & Preparation

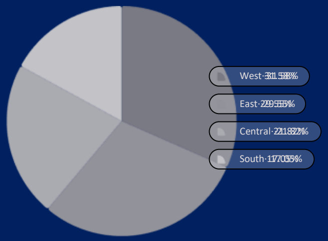
Raw data was systematically cleaned and enriched to ensure accuracy and analytical readiness before any exploratory analysis was performed.

1. Remove Missing Values
2. Standardise Dates
3. Engineer Columns
4. Ready for Analysis

Feature engineering added **Total Sales**, **Year**, and **Month** columns, enabling time-series analysis and aggregated insights across the dataset.

Exploratory Data Analysis





Power BI Dashboard

$2.3M

Total Sales

9,994

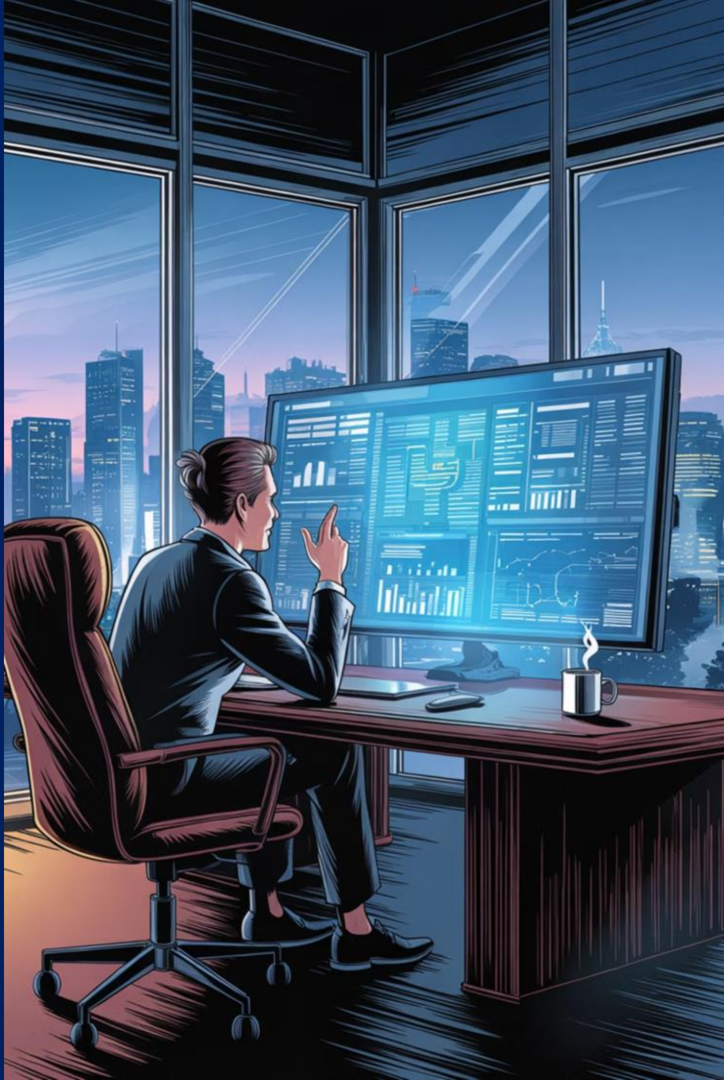
Total Orders

$229

Avg. Sales

Dashboard Features

- Interactive slicers by Region, Category, and Year
- Dynamic filters for drill-down analysis
- KPI cards with real-time metric updates
- Trend lines and comparative bar charts



Key Insights

Technology Leads

Technology category generated the highest sales across all product segments

West Dominates

West region contributed the highest regional revenue share at ~31%

Seasonal Peaks

Sales surged in September, November, and December — clear seasonal patterns

Sub-Category Growth

Top sub-categories like Phones, Chairs, and Binders drove major business volume

Statistical Validation

Hypothesis Testing

An independent samples **t-test** was conducted to evaluate whether the West region's sales performance is statistically distinguishable from other regions.

Result: p-value < 0.05 — the null hypothesis is rejected. West region sales are statistically significantly higher.

1

H_0 — Null Hypothesis

No significant difference exists in sales performance across regions

2

H_1 — Alternate Hypothesis

West region has significantly higher sales than other regions

Business Recommendations

Target High-Performing Regions

Increase marketing investment in the West and East regions to maximise revenue yield

Promote Top Products

Prioritise Technology sub-categories like Phones and Accessories in promotional campaigns

Data-Driven Strategy

Embed Power BI dashboards into routine business reviews for continuous, evidence-based decisions

Lift Underperforming Segments

Investigate and address low-performing categories through targeted promotions and inventory adjustments

Thank You

Data analysis drives smarter decisions. EDA, dashboarding, and statistical storytelling together reveal the growth opportunities hidden within retail data.

Insights Delivered	Tools Used
EDA, statistics & dashboards	Python · Power BI · Statistics

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